OPTIMA ACADEMY ONLINE · FACILITATOR GUIDE

Teacher Manual

M/J Grade 7 Digital Discoveries · Course code **DD-7** · Florida CPALMS #0200022

FACILITATOR COPY

This manual contains answer keys, rubrics, grading cut-offs, free-response answer guides, standards codes, and sensitive-content notes.

Do not publish to students. Keep it in the teacher/staff space only.

Full-year, fully online and asynchronous course (~38 instructional weeks). Built to be imported into Canvas; interactive lessons are hosted on GitHub Pages and embedded in Canvas pages. This document synthesizes the finished, built course as it exists in the repository; items that could not be verified from a built artifact are marked VERIFY. Prepared 2026-07-08.

How to use this manual

- Read **Sections 1–2** first: they tell you what a student actually sees and does, and where the graded touchpoints are.
- **Sections 5–7** are your day-to-day grading references (gradebook setup, every rubric, and a per-item "how to grade this one" catalog). Keep them open in SpeedGrader.
- **Sections 9–10** help you run the week and plan your time; the workload graphic shows where the heavy weeks fall.
- **Sections 11–13** are for audit, culture, and family communication.

> **One structural note before you begin.** The Optima *optima-canvas-assignments* tier model (Tiers 1–4, a 60/25/15 weighting, and survey-quiz module wrappers) was **not** the design used to build this course. Digital Discoveries uses a simpler, fully-online structure: **two weighted gradebook groups — Asynchronous/Independent Work at 70% and Projects & Capstone Portfolio at 30%**, with unit *Overview pages* rather than survey-quiz wrappers. This manual documents the course as built. Where the tier vocabulary would normally appear, the real 70/30 structure is described instead. If the school later wants this course re-aligned to the tier model, that is a build change, not a manual change. VERIFY with the curriculum lead if tier alignment is required.

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1 How this course works

Digital Discoveries is a year-long classical computer-science course framed by the Trivium (Grammar → Logic → Rhetoric) and an expedition through the canon of computing thinkers — al-Khwārizmī, Boole, Lovelace, Turing, Hopper, Berners-Lee, and others. It is **fully online and asynchronous**: no live classes, no synchronous seminars. Students move through interactive lessons, practice, discussion boards, quizzes, and projects on their own schedule.

1.1 A typical lesson

Every lesson is a single Canvas *Page* that embeds the interactive lesson (hosted on GitHub Pages) inline. Walking one real lesson top to bottom — **Unit 1, Lesson 2: Hardware & Software** — a student sees, in order:

1. **A "Skip to main content" link** and a lesson header: a kicker ("Unit 1 · Lesson 2"), the title, and an **essential question** that frames the reading.
2. **Content blocks** under clear `<h2>` headings, with key terms marked as vocabulary and short "Key Words" boxes. Grade-7 reading level, classical voice.
3. **An embedded interactive** — here, the *Inside the Machine* explorer (parts, boot-order stepper, sort self-test). The student does the hands-on work right inside the page. It gives immediate feedback and produces *no submission*.
4. **Self-checks** — a few comprehension questions with instant feedback.
5. **A virtue reflection prompt** woven into the close of the unit's lessons (responsibility, courtesy, perseverance, honesty, and so on).
6. **The graded touchpoint:** right after the lesson page, the module lists that lesson's **Assignment** (a short text-entry / upload / media task, 10 points). In Unit 1 Lesson 2 that is "Inside the Machine." This is the only part a teacher grades.

> Rule of thumb: **the interactive is practice; the interleaved Assignment is the grade.** A module lists Overview → Lesson page → that lesson's Assignment(s) → next Lesson … → Unit Quiz.

1.2 A typical week

The course runs about **38 instructional weeks**, roughly one lesson-topic per week, with a unit spanning 3–5 weeks (see `teacher/pacing-guide.csv`). A normal week for a student looks like:

Segment	What the student does	Graded?	Group
Read & explore	Work through the lesson page and its embedded interactive; do the self-checks	No (formative)	—

Lesson assignment	Short text-entry / upload / media response tied to that lesson (10 pts)	Yes	Async 70%
Discussion (unit-level)	Post to the unit's anchor-text discussion board and reply to a peer (10 pts)	Yes	Async 70%
VR build (once per unit)	Build a small space and record a 1–2 min walkthrough; no-VR equivalent available (10 pts)	Yes	Async 70%
Unit quiz (end of unit)	One auto-graded quiz; some units add one free-response item	Yes	Async 70%
Unit project (U7, U9, U10)	A larger build / investigation / portfolio with a recorded walkthrough	Yes	Projects 30%

Estimated student time is roughly 3–4 hours per week; short/holiday weeks are marked in the pacing guide.

1.3 Major assessments

Every unit ends with a quiz; three units carry a major project. The projects and the capstone are the entire 30% Projects group.

Assessment	Unit	Group	What it asks	Approx. grading time / student
Unit quizzes (×10)	All	Async	Auto-graded recall/reasoning; 7 units add one free-response item	~0 auto; +2–3 min for the free-response item
Unit 7 Build Project (Scratch)	7	Projects (20 pts)	Loop + conditional + variable + expression + a deliberate non-iterative section; recorded walkthrough	~6–8 min
Unit 9 Simulation Investigation	9	Projects (20 pts)	State a hypothesis, modify a simulation fairly, report an honest conclusion; recorded walkthrough	~6–8 min
Capstone Portfolio	10	Projects (30 pts)	Published portfolio (4–6 artifacts) + 2–4 min recorded presentation + final reflection	~12–15 min

The free-response quiz items live in Units 1–6 and 8. Units 7, 9, and 10 have no essay item because the project/capstone is that unit's constructed response.

2 Ungraded interactives & unit overviews

2.1 The ungraded interactives

Every lesson embeds a hands-on interactive (sorters, steppers, sandboxes, matchers, simulations) served from GitHub Pages and rendered inline in the Canvas page. These:

- give the student **immediate feedback** and are **keyboard-operable, screen-reader-labeled, and store no student data**;
- produce **no submission and need no grading** — they cost you nothing;
- are **formation by practice, not busywork to chase.** A student who skips one does not create a missing grade; the effect shows up indirectly, in weaker quiz or assignment performance on that skill.

You do not monitor completion of the interactives directly. If a student's quiz/assignment on a skill is weak, that is your signal to point them back to the relevant interactive for another rep.

2.2 Unit Overview pages (not survey-quiz wrappers)

> **Where this course differs from the standard Optima design:** it does **not** use Canvas Quizzes in survey mode as intro/outro "wrappers." There is no first/refined essential-question capture, confidence rating, or virtue-reflection survey collected as sortable cohort data. Instead, each unit opens with an **Overview page** (a normal Canvas Page), each lesson carries its **essential question** in the reading, and virtue reflection is woven into lessons and assignments. This is noted honestly so you do not go looking for wrapper analytics that were never built.

Your highest-leverage early-warning move, adapted to this build

Because there is no outro survey to triage, the equivalent move is to **read each unit's Lesson 1 discussion board and first lesson assignment early in the unit** (about 15–20 minutes). These are your first real read on the cohort:

- **Watch for:** repeated misconceptions in the discussion, students who did not post, thin or copied first assignments, and confusion clusters around one concept.
- **Act on it:** a targeted class announcement (Section 9), an individual warm check-in, or an Ignite AI-drafted nudge (Section 4).

> VERIFY If you want true cohort analytics, you can add a short ungraded Canvas survey-quiz at each unit's start/end (essential question + confidence + "what's still unclear"). That is an optional enhancement, not part of the built course.

3 AI & VR in this course (student-facing)

3.1 AI

The built course treats AI as an **object of study, not a required tool.** Its posture, drawn from the actual lessons:

- **Where AI appears:** Unit 6 (The Universal Machine) anchors on Alan Turing's 1950 paper "Computing Machinery and Intelligence," asking "can machines think?" and introducing the imitation game. AI is also named lightly on the course home page and syllabus as part of the field's story.
- **No AI tool is required** of students anywhere in the course, and there are **no separate "AI-awareness" callout boxes** in the lessons. When you speak to AI with students, do it through the Turing material: what the imitation game does and does not settle, and the difference between a machine following instructions and a machine "thinking."
- **Do not add new AI requirements.** Keep any AI discussion factual, not persuasive, and never tied to a specific product.

Note the clean separation: the section below (4) is about *your* AI tooling as the teacher. That is distinct from what students are asked to do, which is the picture above.

3.2 VR

Each unit includes **one student-built VR activity** (10 pts, in the Async group). The pattern is identical across all ten units:

- The student **builds a small space themselves** that represents the unit's core idea (for example, a machine with labeled hardware and software layers in Unit 1; a "Logic-Gate Room" in Unit 3; a "Trace Wall" in Unit 6; a "Portfolio Gallery" in Unit 10). There is nothing pre-made to "join."
- They **record a 1–2 minute walkthrough** explaining each label in their own words, and submit the recording (or a link/upload).
- A **no-VR equivalent** is always provided (a labeled slide or short doc + one explanatory sentence), graded identically, so every student can complete it.
- It is deliberately **short (~20–30 minutes)** and graded on correctness of labels and clarity of explanation, not on production polish.

How you review them: watch the short recordings (or read the no-VR docs) and grade against the VR build's stated criteria. A "gallery walk" works well — a discussion thread where students post a still or link and one classmate replies — but that is optional.

> **Platform note.** Optima is moving to a new VR platform, so this manual describes VR by *what the student does*, not by product. The built assignment pages currently name a specific tool ("Arthur");

> when the new platform is confirmed, update the tool name in `assessments/unit-*/assignments/unit-vr-*.html`. The task and grading do not change. **VERIFY** the current platform before the year begins.

4 Using Ignite AI to support students

> **VERIFY** Ignite AI's exact feature names and availability vary by Canvas release and by what Optima has enabled. Confirm the current feature set against Optima's Canvas instance before relying on any specific button. The capabilities below are described functionally.

With roughly 500 asynchronous students, your scarce resource is attention. Canvas's built-in AI assistant (Ignite AI) exists to help you make a 500:1 ratio feel personal — not to grade for you, and never to watch students. The standing rule under everything below is: **AI drafts, the teacher decides.** You read, edit, and own anything sent to a student or family.

4.1 Drafting individualized feedback

For project and free-response grading (the time-consuming Tier of work), let Ignite AI draft feedback from the rubric and the student's submission, then **edit it into your own voice** before posting. Same human judgment, faster turnaround. Always lead with what is working and name the next step (Section 8).

4.2 Summarizing discussions & quiz results

Ask Ignite AI to summarize a unit's discussion board or the quiz item-analysis to surface the cohort's sticking points quickly — for example, "which Unit 4 quiz items did students miss most, and what misconception does that suggest?" Use the summary to aim your weekly announcement and any re-teaching.

4.3 Surfacing students who need a nudge

Use analytics plus Ignite AI to **find** students early — missing the Lesson 1 discussion, declining quiz scores, a stalled project — and to draft a warm, specific check-in message. This is care, not surveillance: the goal is a hand extended before a student falls behind, never a case built against them.

4.4 Generating practice variants

For a student who needs another rep on a skill (order of operations, tracing, loop prediction), ask Ignite AI to generate a fresh practice variant modeled on the relevant lesson interactive. Point the student back to the ungraded interactive first; the variant is extra practice, not a new graded item.

4.5 Communicating with families

Ignite AI can draft clear, kind progress summaries for families and translate messages when helpful. Edit for accuracy and tone, then send. For sensitive-content notices, use the scripts in

Section 13 as your base.

> **Governing rule for this whole section:** every use above is *coach, not catch*. AI is never used to rank, surveil, or auto-penalize. It drafts; you decide and own the result.

5 Grading categories & Canvas gradebook setup

5.1 The two weighted groups

The course ships with two assignment groups and **group weighting turned on** (`apply_assignment_group_weights=true`). If you build the course fresh instead of importing, reproduce exactly this:

Assignment group	Weight	What goes in it	Grading method
Asynchronous / Independent Work	**70%**	All lesson assignments, all discussion boards, all VR builds, and **all 10 unit quizzes**	Auto (quizzes) + rubric/points (assignments)
Projects & Capstone Portfolio	**30%**	Unit 7 Build Project (20), Unit 9 Simulation Investigation (20), Capstone Portfolio (30)	Rubric

> There is no separate synchronous group — this is a fully-online course, so the share that a live component would carry was folded into the Async group. This is the confirmed 70/30 scheme.

5.2 Step-by-step Canvas configuration

1. **Import the package first** (`dist/oao-digital-discoveries-gr7.imscc`) — groups, weights, quizzes, and assignments arrive configured. The steps below are for verifying or rebuilding.
2. **Create the two assignment groups** named exactly as above (Gradebook → Settings → Assignment Groups).
3. **Turn on weighting:** Gradebook → Settings → "Weight final grade based on assignment groups." Set 70% and 30%.
4. **Set the grading scheme** to your school's standard letter scale (the package leaves `grading_standard_enabled=false` ; enable and select the OAO scheme if required). VERIFY the exact scale with the registrar.
5. **Confirm quiz settings** (they import this way): quiz type = graded assignment, **3 attempts, keep highest**, **shuffle answers on**, show correct answers on. Quizzes sit in the Async group.
6. **Confirm assignment submission types:** assignments accept **online text entry, file upload, or media recording** — this is what lets students type, upload, or record the VR/project walkthroughs.
7. **Late / missing policy:** set a course-wide policy that fits an async cohort (a modest per-day deduction with a generous window works well). This course has no built-in due dates —

you set the pacing dates when you deploy.

5.3 Gradebook map

Every graded item, by unit. All 10-point items and all quizzes are in the Async group; only the three project rows are in the Projects group.

Unit	Async assignments (10 pts each)	Unit quiz (points)	Projects group
1 — The Machine & the Human	6 (3 activities, Engelbart discussion, adaptive-tech match, VR build)	10 (7 auto + 1 free-response, 3)	—
2 — The Honest Collaborator	6 (4 activities, well-being discussion, VR build)	13 (10 auto + 1 free-response, 3)	—
3 — The Laws of Thought	4 (Boole discussion, 2 activities, VR build)	11 (8 auto + 1 free-response, 3)	—
4 — Order & Operation	4 (order discussion, 2 activities, VR build)	11 (8 auto + 1 free-response, 3)	—
5 — The First Programmer	4 (Lovelace discussion, 2 activities, VR build)	12 (9 auto + 1 free-response, 3)	—
6 — The Universal Machine	5 (Turing discussion, 3 activities, VR build)	12 (9 auto + 1 free-response, 3)	—
7 — Patterns & Repetition	5 (patterns discussion, 3 activities, VR build)	9 (9 auto, no essay)	**Build Project — 20**
8 — The First Bug & the Life Cycle	4 (first-bug discussion, 2 activities, VR build)	12 (9 auto + 1 free-response, 3)	—
9 — Worlds in Miniature	3 (models discussion, 1 activity, VR build)	9 (9 auto, no essay)	**Simulation Investigation — 20**
10 — Weaving the Web (Capstone)	4 (proposal discussion, practice page, peer-feedback discussion, VR build)	9 (9 auto, no essay)	**Capstone Portfolio — 30**

Totals: 48 assignments + 10 quizzes = 58 graded items. Projects group = 20 + 20 + 30 = 70 points. Within a weighted group, Canvas normalizes by points, so the differing raw totals per unit do not distort the final grade.

> **Module-progress view.** Turn on Modules → requirements (mark-as-done / must-submit) to get an at-a-glance completion signal across the ungraded scaffolding, since the interactives themselves report nothing.

6 Rubrics & free-response answer guides

Paste each rubric into its Canvas item as a **native Canvas Rubric** so it scores in SpeedGrader. The free-response answer guides below are for you only — never show them to students.

6.1 The recurring rubric shape

Every teacher-graded written/project item uses **four criteria on a 4-level scale (16 points, scaled to the item's gradebook value)**. The fourth criterion is always an integrity/clarity criterion, and the same standing rule applies everywhere: **a submission that is not a sincere attempt (nonsense, off-topic, or not the student's own words) earns no credit.** Keeping the shape constant holds grading near ~5 minutes per student and keeps standards steady across units.

6.2 Per-unit constructed-response rubrics

Unit / task	Four criteria (4×4 = 16 pts, scaled)
U1 — Written explanation: how a device's hardware & software work together (5–8 sentences)	Hardware/software relationship · Concrete examples (part, software, interaction) · Clarity & completeness · Honesty & care
U2 — Career + ethics reflection (5–8 sentences, cite source)	Career analysis · Ethics connection · Citation · Clarity & honesty
U3 — Design a Boolean rule (≥2 of and/or/not)	Correct use of and/or/not · Rule fits the intent · Explanation (kept & excluded) · Clarity & honesty
U4 — Develop two expressions (arithmetic + logical, with parentheses)	Arithmetic expression · Logical expression · Precedence understanding · Clarity & honesty
U5 — Write & predict (iterate a list; name types; predict output)	Iteration · Variables & types · Indexing (0/1 base) · Output predictions
U6 — Trace & report (Euclid GCD, a=24, b=9)	Trace table · Final output · Plain-words explanation · Error-type judgment
U7 — Build Project (Scratch; Projects group, 20 pts)	Required elements (loop/conditional/variable/expression) · Iterative + non-iterative · Program quality · Walkthrough & bug story
U8 — Repair report (three-bug program)	Bugs found & typed · Fixes · SDLC phase · Class definition
U9 — Simulation Investigation (Projects group, 20 pts)	Hypothesis · Fair modification · Data & findings · Honest conclusion

U10 — Capstone Portfolio (Projects group, 30 pts)	Curation · Craft & accessibility · Recorded presentation · Honesty & credit

Full 4-level descriptors for each criterion live in `assessments/unit-NN/rubric.md`. For U7, U9, and U10 the 16-point rubric scales to the item's 20 / 20 / 30 gradebook points. A refuted hypothesis honestly reported (U9) scores as well as a supported one.

6.3 Free-response quiz answer guides (teacher only)

Seven unit quizzes end with one manually-graded free-response item worth **3 points**. Units 7, 9, and 10 have **no** essay item — the project/capstone is the constructed response. Score each guide out of 3: full / partial / none.

Unit	What a full-credit answer contains	Partial / none
U1 (Q8)	Names a device; names ≥1 hardware part, ≥1 software, and ≥1 way they interact; explains that each needs the other.	Partial: names parts but not the relationship. None: no real hardware/software link.
U2 (Q11)	Describes a tech career (what they do + preparation), links one responsible/ethical habit to that job, and cites the source of the career info.	Partial: career or ethics link present but not both, or no citation. None: off-topic.
U3 (Q9)	A Boolean rule using ≥2 of and/or/not correctly, that matches the stated intent, with what it keeps and excludes explained.	Partial: one operator or vague explanation. None: operators misused.
U4 (Q9)	An arithmetic expression (3 numbers, 2 operators, one pair of parentheses) evaluated step-by-step in precedence order, plus a logical expression (comparison + AND/OR + parentheses) with its truth value.	Partial: one of the two expressions sound. None: precedence not shown.
U5 (Q10)	A code segment that iterates a list; variables named with correct types; the indexing base (0 or 1) stated; exact predicted output for two different input lists.	Partial: loop works but predictions or types shaky. None: no real iteration.
U6 (Q10)	Full trace for a=24, b=9: (24,9)→r=6→(9,6); (9,6)→r=3→(6,3); (6,3)→r=0→(3,0); **output = 3**; each loop line explained.	Partial: correct answer, incomplete table. None: no usable trace.
U8 (Q10)	Line 4 = **logical** (− should be +); Line 5 = **runtime** (count is 0 → ÷0; set count=3 or guard it); Line 6 = **syntax** (unclosed quote). Bug-catching phase = **Test**. Class defined in one sentence.	Partial: two bugs typed correctly. None: one or none.

7 Discussions, quizzes & activities: how to run & grade

7.1 By type (general guidance)

Type	What it is & how it's scored	Recommended move
Auto-graded quiz items	Multiple-choice / true-false; scored instantly; 3 attempts, keep highest, answers shuffled	Do not grade. Read the item-analysis for cohort misconceptions.
Free-response quiz item	One per quiz in U1–6, 8; 3 pts; manual	Score full/partial/none from the answer guide (6.3). Batch them.
Discussion board	Anchor-text post + one peer reply; 10 pts	Batch-read on a simple substantive / thin / missing scale; drop 2–3 public model replies that name good thinking.
Lesson activity assignment	Short text/upload response tied to a lesson; 10 pts	Quick points check against the task's stated "graded on…" line; brief feedback.
VR build	Student-built space + recorded walkthrough (or no-VR doc); 10 pts	Watch/read; grade labels + clarity, not polish. Optional gallery-walk thread.
Project / Capstone	Scratch build, simulation investigation, portfolio + recording; Projects group	Grade with the unit rubric; let Ignite AI draft feedback, then edit and own it.
Embedded interactive	In-lesson practice; instant feedback	Not graded, no submission. Redirect struggling students here for another rep.

7.2 Per-instance quick reference

The unit's **discussion** is almost always Lesson 1 (the anchor-text response); the **free-response** shape is in 6.3; the **project** rubric is in 6.2.

Unit	Anchor-text discussion (L1)	Notable graded items
1	Engelbart "Mother of All Demos" (posted at L4)	Build-an-algorithm; Inside-the-Machine; adaptive-tech; VR Build-the-Machine; free-response Q8
2	Well-being / "person behind the screen" (L4)	Collaboration tools; team roles; source evaluation; ethics & careers; VR Team Workspace; free-response Q11
3	Boole, *Laws of Thought*	Truth tables; Boolean rule; VR Logic-Gate Room; free-response Q9 (design a rule)
4	al-Khwārizmī / order of operations	Arithmetic & logical expressions; VR Order-of-Operations Gallery; free-response Q9 (two expressions)

5	Lovelace, *Notes*	Indexing; write-a-loop; VR Memory Store; free-response Q10 (write & predict)
6	Turing, "Computing Machinery and Intelligence"	Trace Euclid; predict sections; error types; VR Trace Wall; free-response Q10 (trace, answer = 3)
7	Patterns / Fibonacci	Write-a-loop; two-ways; Scratch explore; VR Fibonacci Garden; **Build Project (20)**
8	Hopper "first bug" logbook	SDLC; design-a-class; VR Software Workshop; free-response Q10 (repair report, 3 bugs)
9	Models & simulations	Tweak report; VR Model Museum; **Simulation Investigation (20)**
10	Berners-Lee "Information Management: A Proposal"	Practice page; peer-feedback round; VR Portfolio Gallery; **Capstone Portfolio (30)**

8 Coach, not catch

Your job is to move each student forward, not to detect and punish. In practice:

- **Feedback names the next step, not just the error.** Lead with what is working, then the one thing to try next.
- **Allow re-attempts on practice.** Quizzes already keep the highest of three attempts; treat the lesson interactives as unlimited practice. Practice is a learning tool, not a trap.
- **Use analytics and Ignite AI to find students who need help early,** then reach out — not to build a case against them.
- **On suspected dishonesty or AI misuse:** open a conversation. Ask the student to talk through their thinking. Treat it as formation. Avoid accusation-first workflows. The rubrics' "not a sincere attempt earns no credit" line is about genuine effort, applied with a conversation, not a trap.
- **Public discussion comments model good thinking;** they do not single out weak work for criticism. Praise in public, correct in private.

9 Weekly announcement scripts

In an async course the weekly announcement is your main presence. Keep it short and human. Fill the bracketed slots when you post.

9.1 Reusable template

Subject: [Unit N] — this week in Digital Discoveries

Hi everyone — this week you should be in **[Unit N, Lesson X]**. The focus is **[one-line focus]**.

Don't miss: [the one due item or tricky concept]. Due **[date]**.

One tip: [a single concrete pointer — "trace it by hand before you predict," "keep your Scratch project small and finished"].

You've got this. Reply here or message me if anything is unclear. — [Your name]

9.2 Drafted scripts, one per unit

Unit	Focus & the one thing not to miss
1	What computing is; hardware vs. software; how we meet the machine. **Don't miss:** the Engelbart discussion — post *and* reply to a classmate.
2	Collaboration, honest research, and the person behind the screen. **Don't miss:** the well-being discussion; and cite your source on the career reflection.
3	Boolean logic — and / or / not, and truth tables. **Tip:** build the truth table in the sandbox before you write your rule.
4	Order of operations in arithmetic and logical expressions. **Tip:** parentheses first, then ×÷, then +− — show every step.
5	Variables, lists, indexing, and iterating a list. **Don't miss:** state whether your list is 0-based or 1-based and predict the exact output.
6	Program tracing (Euclid's GCD) and the three error types. **Tip:** the written trace for a=24, b=9 lands on 3 — do it by hand.
7	Patterns, loops, and the Build Project. **Don't miss:** the Scratch project (Projects grade) — small and finished beats big and broken.
8	Debugging, the software life cycle, and classes. **Tip:** for the repair report, name each bug's *type* and a fix that truly cures it.
9	Models, simulations, and testing a hypothesis. **Don't miss:** the Simulation Investigation — a refuted hypothesis honestly reported earns full marks.

10	The web, design, and the Capstone Portfolio. **Don't miss:** the portfolio + recorded presentation — start assembling artifacts now.

9.3 Situational scripts

Mid-unit check-in

If this unit's first discussion or quiz felt shaky, that's normal — the ideas build. Redo the lesson interactive [name] for another rep, and message me one specific question. I'll walk you through it.

Pre-project / pre-capstone

[Project/Capstone] is coming up and it's part of your Projects grade. Read the rubric first — it tells you exactly what earns full marks. Keep it small, finished, and honest. Recording tips are on the assignment page.

Re-engagement (quiet student)

Hi [name] — I noticed you haven't posted in [Unit N] yet and I wanted to check in. Nothing's wrong that we can't fix. What's the one thing getting in the way? Even a one-line reply helps me help you.

10 Teacher workflow & workload map

10.1 A normal week, in order

1. **Scan the new unit's L1 discussion + first assignments** (~15–20 min) for confusion and non-starters (your early-warning move, Section 2).
2. **Batch-grade discussions** on the substantive/thin/missing scale; drop 2–3 public model replies.
3. **Grade the week's lesson assignments and any free-response quiz item** against the guides in Section 6.
4. **Grade any project/capstone submissions** with the unit rubric (Ignite AI drafts, you edit).
5. **Post the weekly announcement** (Section 9).
6. **Answer messages** and send any warm check-ins.

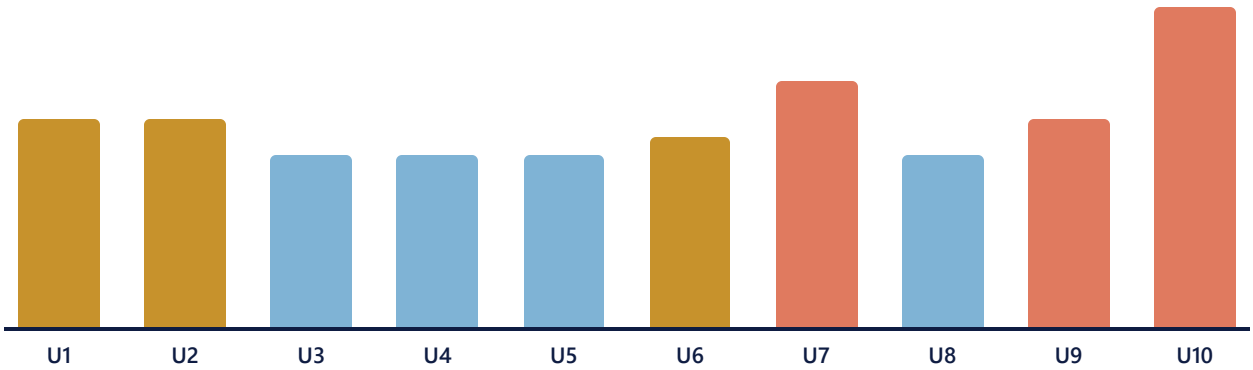
Auto-graded quizzes cost ~0 grading time. The weight sits in discussions, free-response items, and — heavily — the three project modules.

10.2 Heavy modules to plan around

- **Unit 7** — first major project (Scratch builds + recorded walkthroughs to review).
- **Unit 9** — Simulation Investigation (hypothesis + recordings to assess).
- **Unit 10** — the Capstone: portfolios + 2–4 min presentations + reflections. The single heaviest grading push of the year; it also lands in the final weeks, so budget time and start early feedback.
- Units 1, 2, and 6 are moderate (more assignments and/or a free-response item plus an anchor discussion).

10.3 Workload across the year

Relative facilitator-grading load by unit, estimated from the actual tiered item counts (discussions, free-response items, VR builds, and projects). Auto-graded quizzes contribute almost nothing.



Blue = light (activities + one discussion). Gold = moderate (more assignments and/or a free-response item). Coral = heavy (a major project or the capstone). Plan extra time on the coral units, especially Unit 10.

11 Standards coverage map (CPALMS #0200022)

Verified from the course's alignment matrix (`standards/alignment-matrix.csv`). Coverage report: **52 benchmarks tracked (31 SC.7 + 21 embedded); none unmapped, none orphaned.** Codes are teacher-copy only — they never appear on student pages.

11.1 By standard (SC.7 course benchmarks)

Code	Benchmark (verbatim)	Unit(s)	Assessed in
CC.1.1	Apply multimedia tools for local and global group collaboration.	2; 10	U2 & U10 quizzes; Capstone; peer-feedback; VR Team Workspace
CC.1.2	Identify productivity tools for collaboration.	2	U2 quiz; L2.1 assignment
CC.1.3	Identify individual roles within a collaborative team.	2	U2 quiz; L2.2; VR Team Workspace
CO.1.3	Describe the relationship between hardware and software.	1	U1 quiz; L1.2; VR Build-the-Machine
CO.1.4	Utilize a set of websites to find information for a given topic.	2	U2 quiz; L2.3 source-evaluation
ET.3.1	Describe ways in which adaptive technologies can assist users in their daily lives.	1	U1 quiz; L1.4 assignments
ET.3.2	Identify ways humans interact with computers.	1	U1 quiz; L1.3
ET.3.3	Identify ways humans interact with hardware components.	1	U1 quiz; L1.3
HS.1.2	Discuss the impact of online disinhibition on individuals and society.	2	U2 quiz; L2.4 discussion
HS.2.1	Identify the digital practices that may affect your physical and mental well-being.	2	U2 quiz; L2.4 discussion
HS.3.3	Explain the possible consequences of cyberbullying and inappropriate use of digital media and communication on personal life and society.	2	U2 quiz; L2.4 discussion
PE.1.2	Write a code segment that will explore a list using iteration.	5; 7	U5 & U7 quizzes; write-a-loop; U7 project
PE.1.3	Develop a logical expression using operator precedence.	4	U4 quiz; L4.3; VR Order Gallery

PE.1.4	Develop an arithmetic expression using operator precedence.	4	U4 quiz; L4.2; VR Order Gallery
PE.1.5	Identify the types and uses of variables in a program.	5	U5 quiz; Lovelace discussion; VR Memory Store
PE.1.6	Develop problem solutions using a block programming language.	7; 9	U7 & U9 quizzes; U7 project; U9 investigation
PE.1.7	Create online content using advanced design tools.	10	U10 quiz; practice page; Capstone
PE.1.8	Identify different types of programming errors.	6; 8	U6 & U8 quizzes; error-types; first-bug discussion
PE.1.11	Create iterative and non-iterative structures in a code segment.	7	U7 quiz; two-ways; U7 project; VR Fibonacci Garden
PE.2.1	Predict outputs while showing an understanding of inputs.	5; 6	U5 & U6 quizzes; predict-sections
PE.3.1	Define parameters for individual and collaborative projects using Boolean logic.	3	U3 quiz; truth-tables & boolean-rule; VR Logic-Gate Room
PE.3.2	Modify and create a simulation to analyze and illustrate a concept in depth.	9	U9 quiz; tweak report; U9 investigation
PE.3.3	Use modeling and simulations to test scientific hypotheses.	9	U9 quiz; U9 investigation
PE.3.4	Define the concept of a class related to object-oriented programming.	8	U8 quiz; design-a-class
PE.3.5	Identify the purpose of indexing the order of elements in a list.	5	U5 quiz; indexing; VR Memory Store
PE.3.6	Perform program tracing to predict the behavior of programs.	6	U6 quiz + written trace; trace & predict; VR Trace Wall
PE.3.7	Identify the types and uses of variables in a program.	5	U5 quiz
PE.4.1	Define the phases of the software development life cycle.	8	U8 quiz; SDLC; VR Software Workshop
TI.1.2	Evaluate the responsible and irresponsible use of information on collaborative projects.	2	U2 quiz; L2.5
TI.1.4	Analyze technology-related career paths.	2; 10	U2 & U10 quizzes; L2.5; peer-feedback

TI.2.1	Describe legal and ethical behaviors when using information and technology and describe the consequences of misuse.	2	U2 quiz; L2.5

The 21 embedded B.E.S.T. / framework standards (SC.K12.CTR.*, MA.K12.MTR.*, ELA.K12.EE.*, ELD.K12.ELL.MA.1) are fully mapped in `standards/alignment-matrix.csv` / `.html`; the coverage report confirms zero gaps and zero orphans.

11.2 By unit (quick reference)

Unit	SC.7 benchmarks carried
1	CO.1.3, ET.3.1, ET.3.2, ET.3.3
2	CC.1.1, CC.1.2, CC.1.3, CO.1.4, HS.1.2, HS.2.1, HS.3.3, TI.1.2, TI.1.4, TI.2.1
3	PE.3.1
4	PE.1.3, PE.1.4
5	PE.1.5, PE.3.7, PE.3.5, PE.1.2, PE.2.1
6	PE.3.6, PE.2.1, PE.1.8
7	PE.1.11, PE.1.2, PE.1.6
8	PE.1.8, PE.4.1, PE.3.4
9	PE.3.2, PE.3.3, PE.1.6
10	PE.1.7, CC.1.1, TI.1.4

> **VERIFY** Pacing variance: CPALMS lists course #0200022 as one semester; OAO delivers it as a full year (~38 weeks) for deeper treatment. The registrar/authorizer sign-off on how full-year delivery is coded on the master schedule and transcript remains open.

12 What makes this course classical & cultivating culture

12.1 What makes it classical

- **The Trivium as the year's spine.** Grammar (foundations & vocabulary, U1–2), Logic (reasoning, Boolean logic, algorithms, tracing, U3–6), Rhetoric (expression & building, U7–10). Students see computing as ordered thought, not button-pushing.
- **Primary-source encounter.** Each unit opens on a real anchor text or artifact — Engelbart, Bush, Boole, al-Khwārizmī, Lovelace, Turing, Hopper, Berners-Lee — met directly through a reading guide and an anchor discussion, joining the Great Conversation of the field.
- **Wonder before mechanism.** Lessons lead with an essential question and the human story, then earn the mechanism — the origin of "algorithm," the first recorded "bug," a 1989 proposal that became the web.
- **The virtue thread.** Responsibility, courtesy, perseverance, honesty, and service recur as genuine reflection prompts, and honesty is a graded criterion in every rubric — "good, true, and beautiful," honestly presented.

12.2 Cultivating culture asynchronously

- Write **substantive public discussion replies** that name a student's good thinking by name — this is how a community forms without a live room.
- Let your **weekly announcement model curiosity**, not just logistics: a sentence of real interest in the week's thinker goes a long way.
- Keep the **virtue thread visible** — reference it in feedback and announcements, not only where the lesson prompts it.
- Treat **feedback as relationship.** A specific, warm, next-step comment is the async equivalent of a good classroom conversation.

13 Sensitive-content flags & family/student scripts

The course is written for ages ~12–13 in a classical, respectful voice. One lesson handles material that families may want advance notice of. (Note: Unit 6 studies Turing's ideas about machine intelligence; the built lesson does **not** discuss his personal persecution or death, so no flag is raised there.)

13.1 The one flagged lesson

Where	What it is
Unit 2, Lesson 4 — "The Person Behind the Screen"	Age-appropriate treatment of **online disinhibition, digital well-being** (screen time, sleep, comparison, effects on mental health), and **cyberbullying and its consequences** (fear, isolation, harm to mental health; harm to a community). Framed around courtesy and the real person on the other side of a message. Standards HS.1.2, HS.2.1, HS.3.3.

13.2 Suggested advance script (edit for your voice and contact info)

Subject: A heads-up about next week's lesson — digital well-being & kindness online

Dear families,

Next week in Digital Discoveries, Unit 2 includes a lesson called "The Person Behind the Screen." It covers healthy technology habits and the real consequences of unkindness online, including cyberbullying. We treat these topics carefully and age-appropriately, and the purpose is practical: to help students recognize that there is a real person on the other side of every message and to build honest, courteous digital habits — a Florida standard for this grade.

If you'd like to preview the lesson or talk it through with your student, I'm glad to help. Please reach out at [your email/contact] with any questions.

Warmly,
[Your name]

If your school requires a formal content-notice process, route this through it. Save any broader debate about online culture for the lesson's own discussion, per house style: give students a neutral setup and let them reason.

Sources. Every fact above is synthesized from the built course: `canvas/` lesson pages, `assessments/unit-*/` (assignments, quiz QTI, answer keys, rubrics), `build/build.py` (gradebook groups, weights, quiz/assignment settings), `standards/alignment-matrix.csv` and `coverage-report.md`, and `teacher/` pacing and facilitation notes. Items marked **VERIFY** depend on Optima's Canvas configuration, the incoming VR platform, or registrar sign-off and should be confirmed before the year begins.

Facilitator copy — do not publish to students.